

EARL LOUIS

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EDUCATION

University of Connecticut, Storrs, CT

Bachelor of Science in Robotics Engineering (Electronics Track) | Expected December 2026

TECHNICAL SKILLS

Competencies: Control Panel Design, Engineering Documentation, Blueprint Reading, PLC Systems, Sensor Integration, Instrumentation, Process Automation, OEE, Industrial IoT, Data Analysis, Lean Manufacturing, Systems Design, Systems Integration, Cross-Functional Coordination, Embedded Systems, Signal Processing, Path Planning, Motion Control, Real-Time Systems, Control Systems, Testing & Validation, Prototyping, 3D CAD, Safety Interlocks

Software & Tools: Ignition (8.1 & 8.3 Credentials), Python, C/C++, AutoCAD, KiCad, LTSpice, ROS, Excel

Protocols: Modbus TCP, EtherNet/IP, MTConnect, OPC UA, I2C, SPI, UART

PROFESSIONAL EXPERIENCE

Barnes Aerospace, Windsor, CT

Control Systems Intern (Smart Connected Factory) | June 2025 to Present

- Integrated CNC machines and legacy assets into a plant IoT network over Modbus TCP and MTConnect for real-time data acquisition, part of a global rollout covering roughly 1000 machines.
- Designed and phased in a plant-wide sensor network using wet-switch and wireless sensors with automated alarm dispatch to response teams.
- Engineered instrumentation and control loops for an upgraded waste tank, authoring the P&ID and control panel for level and temperature sensing, immersion heating, and pump interlocks.
- Integrated a pH probe and flowmeters for continuous environmental monitoring, supporting field testing, troubleshooting, and startup.
- Earned Ignition 8.1 and 8.3 Credentials and built HMI/SCADA displays for real-time data visualization.

Norwich Public Utilities, Norwich, CT

Electrical Engineering Intern | June 2024 to January 2025

- Consolidated five years of load data across 4.8 kV and 13.8 kV networks into filterable analytics workbooks, replacing manual per-request pulls.
- Resolved a residential voltage-drop issue blocking EV charging and delivered redesign layouts for senior engineer review.
- Validated pole-top transformers via turns-ratio testing and flagged out-of-tolerance units for follow-up.

Outlier AI, Remote

Machine Learning Quality Assurance Tester | May 2024 to January 2025

- Reviewed Meta AI responses for accuracy, relevance, and safety, flagging cases that overstepped into medical or mental health advice.

Demers Exposition Services, Inc., Uncasville, CT

Stage Technician | December 2022 to June 2024

- Supported crew leads on load-in and load-out of large arena events, running audio and lighting cable and operating chain hoists during rigging.

FEATURED PROJECTS

Drone Health Monitoring System · FFT Signal Processing, Sensor Integration, Wireless Telemetry, Python

Senior design project. Fitted a quadcopter with current, vibration, and temperature sensors to monitor operational health; onboard Pi ran signal processing and transmitted data over 915 MHz to a ground station dashboard with automated alarms.

Two-Wheeled Path Planning Robot · Path Planning, Computer Vision, Python, C++

Navigation robot with a custom path planner and overhead-camera pipeline for real-time localization.

Sound Intensity Visualizer · Embedded C, Signal Processing, AVR

Audio visualizer showing live waveforms on an LED dot matrix with dB readout on a secondary OLED.

Robotic Can Opening Arm · 3D CAD, Motion Control, Mechanism Design, 3D Printing

Designed and 3D-printed a robotic arm mechanism, modeling all mechanical components in CAD.